Indian Institute of Technology Jodhpur | Department of Mathematics

MAL6150: Probability and Statistics

Practice Sheet: Expectations, Moments, and Their Applications

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Useful formulas. $\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x)$ (discrete), $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$ (continuous), $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

A. Computing expectations

1. The probability mass function of X is

x	0	1	2
$\mathbb{P}(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Find $\mathbb{E}[X]$.

2. Suppose $\mathbb{P}(X = n) = c\,3^{-n}$ for $n = 1, 2, 3, \ldots$, where $c > 0$ is a constant. Determine c, and then compute $\mathbb{E}[X]$.

3. The density of X is $f_X(x) = 2x$ for $0 < x < 1$, and is zero otherwise. Compute $\mathbb{E}[X^2]$.

4. Let X be uniformly distributed on $(-1, 2)$. Compute $\mathbb{E}[|X|]$.

B. Moments and variance

5. Suppose $\mathbb{E}[X] = 3$ and $\mathbb{E}[X^2] = 11$. Compute $\mathrm{Var}(2X - 5)$.

6. Let X be uniformly distributed on $(0, 2)$. Compute the second central moment $\mathbb{E}[(X - 1)^2]$.

C. Distributions of transformed random variables

7. Let $\mathbb{P}(X = -1) = \frac{1}{4}$, $\mathbb{P}(X = 0) = \frac{1}{2}$, and $\mathbb{P}(X = 1) = \frac{1}{4}$. Put $Y = X^2$. Find the probability mass function of Y.

8. Let X be uniformly distributed on $(0, 1)$ and put $Y = 3X + 2$. Find the distribution function $F_Y(y)$.

9. Let X be uniformly distributed on $(-1, 1)$ and put $Y = X^2$. Find the density $f_Y(y)$. (First find $\mathbb{P}(X^2 \le y)$.)

10. A point is chosen uniformly at random with respect to arc length on the unit circle. Let Y be its y-coordinate. Find the distribution function $F_Y(y)$ and density $f_Y(y)$.

D. Applications of the theory of moments

Facts you may use: If $\mathbb{E}|X|^k < \infty$, then $\mathbb{E}|X|^r < \infty$ for $0 < r < k$, and $n^k \mathbb{P}(|X| > n) \to 0$. For $X \geq 0$ and $r > 0$,

$$\mathbb{E}[X^r] = r \int_0^\infty t^{r-1} \mathbb{P}(X > t)\, dt,$$

where the integral may be infinite.

11. Suppose $\mathbb{E}[|X|^3] = 8$. Find $\lim_{n\to\infty} n^3 \mathbb{P}(|X| > n)$.

12. Suppose $X \geq 0$ and $\mathbb{P}(X > t) = (1+t)^{-2}$ for all $t \geq 0$. Using the tail-integral formula, determine whether $\mathbb{E}[X^2]$ is finite.

13. Suppose $\mathbb{P}(|X| > t) \leq 4/t^4$ for every $t \geq 1$. Use the tail-integral formula to show that $\mathbb{E}[X^2] \leq 5$.

14. Suppose $X \geq 0$ and $\mathbb{P}(X > t) = (1+t)^{-3}$ for all $t \geq 0$. Determine all real numbers $r > 0$ for which $\mathbb{E}[X^r]$ is finite.

15. Suppose $X \geq 0$ and

$$\mathbb{P}(X > t) = \frac{1}{(1+t)^2 \log(e+t)}, \qquad t \geq 0.$$

Show that $t^2 \mathbb{P}(X > t) \to 0$ as $t \to \infty$, but $\mathbb{E}[X^2] = \infty$. What does this say about the converse of the finite-moment tail theorem?

16. Suppose X takes values in $\{1, 2, 3, \ldots\}$ and

$$\mathbb{P}(X \geq n) = \frac{1}{n^2}, \qquad n = 1, 2, 3, \ldots.$$

Show that $\mathbb{E}[X]$ is finite but $\mathbb{E}[X^2]$ is infinite.